QP CODE: **Reg. No. :.....**
Name :.....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS)MODEL QUESTION PAPER (THEORY)
SEMESTER FIVE
MG5DSEPHY303– Laser, Non-linear Optics and Fiber Optics
(2024 ADMISSION ONWARDS)

Duration: 1.5 hrs **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse (An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation (Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Answer Type Questions
Answer any **10** questions
Each question carries **2** Mark

1.	Distinguish between a two-level and three-level laser system.	[A][1]
2.	What is the function of a resonator in a laser?	[U][1]
3.	Write down the Schawlow-Townes condition for laser oscillation and explain each term in it.	[K][1]
4.	Sketch the energy level diagram of a He-Ne laser.	[U][1]
5.	Define Q-switching and how it can be achieved?	[U][2]
6.	Compare the intensities of a mode-locked and mode-unlocked laser beam.	[U][2]
7.	What is meant by third harmonic generation?	[U][3]
8.	Explain self-focusing of laser beam.	[U][3]
9.	Distinguish between TE and TM modes in a slab waveguide	[A][4]
10.	Which are the two low-attenuation transmission windows in a silica glass optical fibre. Which unit is normally used to express the attenuation in an optical fibre.	[U][4]
11.	Sketch the schematic of a fibre optic communication system.	[A][4]
12.	Distinguish between step index and graded index optical fibre.	[A][4]

[2×10 = 20]

Part B

Short Essay Type Questions

Answer any **6** questions

Each question carries **5** Mark

13. Obtain the expression for the number of resonant frequencies within an open resonator. [U][1]

14. With the help of a proper schematic diagram and energy level diagram, explain the construction and working of a ruby laser. [A][1]

15. Obtain the condition that must be satisfied to get stimulated emission from a semiconductor laser. [U][1]

16. Explain how Q-switching can be achieved in a laser using Kerr effect. [U][2]

17. What is meant by second harmonic generation? Prove that only non-centrosymmetric crystals can exhibit second harmonic generation. [U][2]

18. A laser operating at a pump wavelength of $\lambda_p = 360$ nm pumps a non-linear crystal to produce parametric generation. If the desired signal wavelength is $\lambda_s = 546$ nm:(1) Calculate the frequency of the pump photon ω_p and signal photon ω_s.(2) Determine the idler frequency and its corresponding wavelength [A][2]

19. Obtain the expression for the characteristic penetration depth of an evanescent wave in an optical fibre. [U][3]

20. A Fiber Bragg Grating is to be fabricated to reflect a target centre wavelength $\lambda_0 = 1300$ nm in a silica fibre core with effective refractive index $n = 1.5$. (i) Calculate the required grating pitch/spacing. (ii) Calculate the effective velocity of this wavelength inside the fibre. [A][3]

[5×6 = 30]